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June 16, 2025

The Honorable Robert F. Kennedy, Jr.
Secretary, Department of Health and Human Services
200 Independence Avenue, SW
Washington, DC 20201

Dear Mr. Secretary,

Clinovations Government + Health (Clinovations) is pleased to submit a response to CMS-0042-NC Request for Information: Health Technology Ecosystem.

The response to the RFI's questions reflects Clinovations' deep experience at the intersection of clinical transformation and digital health. With a strong track record in health IT modernization, interoperability strategy and policy, technical assistance, and data integration, Clinovations supports medical and behavioral health providers, health systems, health information exchanges, and public health agencies in building resilient, connected environments for information exchange, population health management, and stakeholder coordination.

Clinovations' responses are based on its multidisciplinary team's experience modernizing data infrastructure, automating workflows, and transforming health and outcomes. For over a decade, Clinovations has provided digital health support for the Centers for Medicare & Medicaid Services (CMS), the Assistant Secretary for Technology Policy/Office of the National Coordinator for Health Information Technology (ASTP/ONC), the National Institute of Standards and Technology (NIST), the Department of Defense (DoD), the Defense Health Agency (DHA), the Department of Veterans Affairs (VA), and the National Science Foundation (NSF). Our team of clinical informaticists, and specialists in clinical workflow, integration, validation and training, and policy analysis have supported government interoperability initiatives, including the Health IT Certification Program, bringing a perspective that balances the need to certify technology that simplifies provider workflows and reduces documentation burden, while ensuring a standard set of functionality to easily facilitate the exchange and use of interoperable data.

We thank CMS and ASTP/ONC for reviewing our responses and look forward to supporting the government's efforts to modernize and transform healthcare.

Sincerely,

Anita Samarth
Clinovations Government + Health

B. Patients and Caregivers

1. Patient Needs

PC-4. What features are missing from apps you use or that you are aware of today?

a. What apps should exist but do not yet? Why do you believe they do not exist yet?

Mobile Health Apps – Leveraging data collected via mobile health apps. While there are many apps available within the Apple app store, Google Play store, and Certified Health IT App Marketplaces, Clinovations' market analyses and implementation work has found many of these apps, if not the majority, do not support bi-directional data exchange between the third-party mobile app and the Certified Health IT. This represents a silo of collected health data that is currently not exchanged, incorporated, and used to inform a patient's healthcare. Although standards currently exist to support bi-directional data exchange between these systems, such as HL7 FHIR R4, SMART on FHIR, FHIR Write, and FHIR Bulk Data, many of these standards are not widely adopted by all data exchange partners. CMS and ASTP/ONC should promote programs to understand healthcare organizations' current data governance management, structure, and workflows, and advance workflows to review, import, and manage third-party data within Certified Health IT.

Remote Patient Monitoring and Medical Devices – Leveraging data collected from remote patient monitoring and medical devices. Remote patient monitoring and medical devices collect healthcare information, which also represent a silo of collected health data that is currently not exchanged, incorporated, and used to inform a patient's healthcare. For remote patient monitoring and medical devices that saves the collected healthcare information either locally or to the cloud, there needs to be an exchange mechanism for the information to be shared with the Certified Health IT. If the remote patient monitoring device or medical device is accompanied with a third-party mobile app, standards exist for the bi-directional exchange off captured data, although additional work related to data governance and access will need to be conducted. Alternatively, if the remote patient monitoring device or the medical device and the Certified Health IT share the same cloud storage and cloud computing service providers, such as Amazon Web Services, a cloud-based interoperability strategy may be feasible.

PC-6. What features are most important to make digital health products accessible and easy to use for Medicare beneficiaries and caregivers, particularly those with limited prior experience using digital tools and services?

Today, notice of privacy practices are opaque and indecipherable to patients, exposing beneficiaries and caregivers to potential use or sale of their health data without consent. Clinovations has implemented solutions including:

- Education and increasing awareness. Educating caregivers and patients, both publicly and privately insured, and increasing awareness of digital health products will increase knowledge and accessibility of these digital health products. Often, caregivers and patients are unaware of what digital health products are designed for, making it difficult to educate and promote the use

of these products in the first place. Many caregivers and patients may be surprised at the burgeoning market for digital health products.

- Technology accessibility. Some patients who are publicly insured may not have access to technologies, such as a mobile device or a computer/laptop, that are necessary to use digital health products. Additionally, some healthcare sites may not have the resources, whether it's capital or the tools, to provide increased accessibility to the digital health products.
- Plain language data use agreements. Some data use agreements are difficult for caregivers and patients. Data privacy and security agreements should be written in plain language, making it easier for caregivers and patients to understand. By increasing data privacy and security transparency, caregivers are more confident in the digital health product, making it more likely for the product to be implemented in the clinical care workflow.
- Standardized vetting process. Standardizing the vetting and review process for each digital health product would decrease the burden and level of effort to review each digital health product prior to its implementation at a healthcare site. By increasing data privacy and security transparency, caregivers are more confident in the digital health product, making it more likely for the product to be implemented in the clinical care workflow.

PC-7. If CMS were to collect real-world data on digital health products' impact on health outcomes and related costs once they are released into the market, what would be the best means of doing so?

a. ***Define the objective for what real world data will tell CMS about digital health products impact and the data gathering strategy from that single objective, while taking into account the technical architecture, reporting facilitators and barriers, and burden on health products to report.***

CMS may consider collecting real-world data on digital health products to 1) guide its programs and policies; 2) support public health initiatives; and 3) measure value-based care. Using diabetes and glucometers, such as the Abbott Freestyle Libre and its accompanying mobile app, as a model example, CMS could use data collected from glucometers to update its CMS diabetes strategy, report out relevant diabetes metrics and trends, and assess the value and quality of care provided to its diabetes population. While the FDA Sentinel Initiative is already obtaining data from some digital health products, it is used to primarily monitor the safety of FDA-regulated medical products.

Choosing a strategy for collecting the real-world data from the digital health product will depend on factors Clinovations has seen serve as critical implementation factors: 1) the acceptable lag between when data was captured and when the data was exchanged with CMS, 2) cadence for reporting (e.g., single time, monthly, quarterly, annually), and 3) desire for automation.

When data is exchanged in near-real time, repeatedly, and automatically, CMS could consider implementing HL7 FHIR and application programming interfaces (APIs). This would align with other federal agency interest in adopting HL7 FHIR and APIs in the reporting of federal measures. Other federal agency examples include CMS and eCQMs, HRSA and UDS+, CDC and the National Vital Statistics System.

Additional work would need to be conducted to better understand whether the digital health product ecosystem can currently support an API, HL7 FHIR-based data exchange mechanism. For digital health products that do not currently support an API, HL7 FHIR-based date exchange schema, it will be

burdensome to report to CMS. Alternatively, cloud computing, specifically a multi-cloud architecture, could potentially be leveraged, given that a significant percentage of existing digital health products are compatible with cloud computing.

2. Data Access and Integration

PC-8. In your experience, what health data is readily available and valuable to patients or their caregivers or both?

a. What data is valuable, but hard for patients and caregivers, or app developers and other technical vendors, to access for appropriate and valuable use (for example, claims data, clinical data, encounter notes, operative reports, appointment schedules, prices)?

Unstructured and free-text data often contain valuable, critical information, however, it is too often inaccessible (i.e., it is not findable by the caregiver within a note) or not usable, as it is not structured and/or not codified for use in clinical care, analytics, parsing into interoperable data, quality measurement, and other secondary uses of data. This results in duplicative data within Certified Health IT, duplication data collection, and lengthier care experiences.

c. What specific opportunities and challenges exist to improve accessibility, interoperability and integration of clinical data from different sources to enable more meaningful clinical research and generation of actionable evidence?

Standardize ontological standards and data dictionaries. Standardizing data dictionaries between Certified Health IT instances would improve accessibility, interoperability, and integration of clinical data from different sources. Currently, the phrase "if you've seen one EHR, then you've only seen one EHR" rings true, since EHRs are often configured differently from instance to instance to support the care sites' specific needs. Additionally, ontological standards used for each data element may vary from provider to provider, since many ontological standards are recommended and are not required.

Incorporate technologies to leverage unstructured and/or free-text data. Unstructured and free-text data often contain valuable, critical information, however, it is too often inaccessible (i.e., it is not findable by the caregiver within a note) or not usable, as it is not structured and/or not codified for use in clinical care, analytics, parsing into interoperable data, quality measurement, and other secondary uses of data. This results in duplicative data within Certified Health IT, duplication data collection, and lengthier care experiences.

CMS should consider leveraging its regulatory authority to explore the adoption of AI, including ML, NLP, and LLMs, to extract information from free-text and/or unstructured data, codify or tag specific portions of the data, interpret unstructured data (e.g., images, videos).

Create a national provider directory. For HL7 FHIR, a nationwide provider directory of FHIR endpoints does not yet exist. This is critical infrastructure that is needed for the implementation and adoption of HL7 FHIR, since HL7 FHIR endpoints are how external data users access health information within Certified Health IT, using APIs and HL7 FHIR.

Standardize ingestion of external data. While HL7 FHIR supports bi-directional data exchange, organizations who exchange data need additional guidance for assessing and accepting external health data (i.e., health data that was not captured by the organization). Currently, there is no standardized workflow for reviewing external data that is pushed to a health IT system (e.g., external health data that is "written" into an EHR). Does the data sit in quarantine? Does the data get automatically ingested?

Increase data sharing participants. In both clinical and research use cases, participants may be unwilling to share their health data with other caregiver (e.g., opting out of HIEs) or researchers (e.g., not consenting to clinical research trials) for a variety of reasons. Reason may include but are not limited to: 1) concerns with the re-sharing of the data and 2) concerns on how the data will be used.

PC-9. Given that the Blue Button 2.0 API only includes basic patient demographic, Medicare coverage, and claims data (Part A, B, D), what additional CMS data sources do developers view as most valuable for inclusion in the API to enable more useful digital products for patients and caretakers?

c. Is there non-CMS data that should be included in the API?

Since US Core is required by CMS under CMS-0057F, payers will be required to support the exchange of clinical data elements. However, CMS could consider leveraging its regulatory authority to align CMS-0057F and future rulemaking in coordination with ASTP/ONC. With the publication of ASTP HTI-1, ASTP is now requiring US Core 6.1.0 AND USCDI V3. In CMS-0057F, CMS is requiring the implementation and adoption of US Core 3.1.1 (USCDI V1). This is important to align because these two versions of US Core are not fully forwards and backwards compatible. Data elements may have been added, removed, merged, or renamed between versions.

C. Providers

1. Digital Health Apps

PR-2. What are obstacles that prevent development, deployment, or effective utilization of the most useful and innovative applications for physician workflows, such as quality measurement reporting, clinical documentation, and billing tasks? How could these obstacles be mitigated?

Appetite for health IT adoption. Many caregivers are hesitant to switch to new systems and digital health products, since they prefer the familiarity of the old systems and digital health products. Additionally, resources to implement and adopt new technologies may be limited.

Dearth of health IT expertise. While digital health products have been gradually adopted into physician workflows, many providers do not have the resources to fund a fully staffed health IT department, reducing support provided to caregivers who are using these digital health products, time to clean and analyze collected data, and ability to explore new digital health products.

Lengthy workflows and non-intuitive UI/UX. Decreasing the number of clicks and screens within digital health products may improve adoption within physician workflows. A main complaint with current EHRs are the many clicks and screens. This feedback should be carried over into all digital health products.

Incorporation of external data. Currently, there is no standardized workflow for reviewing external data that is pushed to a health IT system (e.g., external health data that is "written" into an EHR). Does the data sit in quarantine? Does the data get automatically ingested? By providing clear guidance on how data should be incorporated into Certified Health IT, providers may be more willing to import data into their EHRs.

Review of digital health products. Currently, it is a lengthy and burdensome process for providers to vet all requested digital health products. A provider organization will not implement or adopt a digital health product if it has not been assessed and reviewed. By standardizing digital health product assessments and review, this would decrease the time providers need to spend to vet each and every digital health product. Alternatively, a central registry of vetted digital health products may be useful, as well. Currently, there are disparate app marketplaces, such as the Apple App store, the Google Play store, and app marketplaces hosted by Certified Health IT.

PR-4. What changes or improvements to standards or policies might be needed for patients' third-party digital products to have access to administrative workflows, such as auto-populating intake forms, viewing provider information and schedules, and making and modifying an appointment?

Standardized incorporation of external data. Although technologies exist to support bi-directional data exchange, there is no standardized workflow for reviewing external data that is pushed to a health IT system (e.g., external health data that is "written" into an EHR). Does the data sit in quarantine? Does the data get automatically ingested? By providing clear guidance on how data should be incorporated into Certified Health IT, providers may be more willing to import data into their EHRs.

Design HL7 FHIR Write IGs by use case and/or workflow instead of by HL7 FHIR resource. In alignment with the prior bullet, provider organizations need guidance on assessing and reviewing external data prior to ingesting it into their EHR. Often times, data is ingested based on clinical use case and/or workflow, not on the specific data element itself. By standardizing this workflow, it provides a standard baseline from which providers may further configure to fit their specific needs.

2. Data Exchange

PR-7. What strategies can CMS implement to support providers in making high-quality, timely, and comprehensive healthcare data available for interoperability in the digital product ecosystem? How can the burden of increasing data availability and sharing be mitigated for providers? Are there ways that workflows or metrics that providers are already motivated to optimize for that could be reused for, or combined with, efforts needed to support interoperability?

The HITECH Act, Meaningful Use, and the Promoting Interoperability Program were instrumental in establishing a standards-based foundation for interoperable health IT. These programs enabled core exchange capabilities and catalyzed EHR adoption. However, the health IT and digital health ecosystem has now evolved well beyond EHR data alone.

Current CMS and federal strategies that concentrate regulatory requirements on specific product types—particularly EHRs—risk limiting data availability and disproportionately burdening a single technology sector. Most hospitals and health systems operate tens to hundreds of IT systems, many of

which are not subject to "open access" mandates. We do not recommend extending regulatory requirements to every system. Instead, we endorse a strategy that prioritizes APIs and data exchange across systems, enabling organizations to deploy digital tools that support clinical and administrative data access, regardless of the originating IT platform.

Today's quality measures and interoperability standards inadvertently constrain data availability to only what is explicitly defined. For instance, while rules require the exchange of electronic health information (EHI), they often exclude comprehensive clinical documentation within the EHR—such as unstructured text, narrative notes, and native document formats. As a result, modern tools that rely on advanced data processing techniques face challenges accessing the full scope of EHR content.

CMS should consider leveraging its regulatory authority to mandate that all clinical documentation in EHRs and certified health IT—including nursing notes, consults, flowsheets, and procedure documentation—be accessible via APIs for patient- and provider-facing applications. I could use its regulatory authority to ensure that all and any clinical documentation within an EHR or "Certified Health IT" is available for patient and provider "app ecosystem" use. CMS should also explore strategies to facilitate access to relevant data stored in non-EHR systems such as radiology, cardiology, oncology, and OB monitoring platforms.

With the rapid advancement of NLP, large language models (LLMs), and AI, the emphasis should shift from prescribing rigid formats to ensuring complete and timely access to source data. APIs that support extraction of the complete Legal Medical Record will promote improved care coordination, analytics, population health management, and clinical research. This shift will help ensure that data flows where and when it is needed—without confining innovation to a specific product class or standard.

Some potential strategies and levers to consider could include:

- Issue guidance or clarify that "EHI" under 45 CFR 171 includes all clinically relevant narrative content by any clinical staff, not just structured fields and not just signed physician notes.
- Enhance §170.315(b)(10) (EHI Export) and address limitations that support API-based exports over static exports.
- Consider interoperability program measures that use longitudinal data by using APIs to access data in other EHRs, HIEs, etc.

PR-8. What are ways CMS or partners can help with simplifying clinical quality data responsibilities of providers?

a. What would be the benefits and downsides of using Bulk FHIR data exports from EHRs to CMS to simplify clinical quality data submissions? Can CMS reduce the burden on providers by performing quality metrics calculations leveraging Bulk FHIR data exports?

CMS and its partners could consider exploring levers to leverage FHIR Bulk Data to pull data for clinical quality measures reporting. This would enable CMS and partners to directly request data from providers. This would also provide additional opportunities in the future for CMS and partners to explore the automation of clinical quality measure reporting.

CMS can reduce provider burden by performing quality metrics calculations leveraging Bulk FHIR data exports. CMS could potentially develop an application that calculates quality metric, if it does not yet exist, using the metric logic provided by CMS eCQMs. In this model, providers would deliver the raw data, assuming it is formatted correctly in a flat file (e.g., CSV). If CMS or partners would like to automate this workflow, CMS and partners could consider the development of APIs and supporting infrastructure (e.g., a data repository), so that the data can be automatically pulled from providers on an agreed upon periodic basis (e.g., monthly, quarterly).

The potential disadvantage of leveraging FHIR Bulk Data is that both the provider and CMS would need to be HL7 FHIR-enabled. It is known that not all providers have adopted and enabled HL7 FHIR. Additionally, HL7 FHIR Bulk Data may not support the exchange of all data required by CMS clinical quality measures. CMS should conduct an analysis to verify FHIR Bulk Data may support the exchange of all data elements required by CMS eCQMs. Lastly, FHIR Bulk Data can be resource intensive, especially for large data pulls. CMS should conduct an analysis to determine if the necessary IT infrastructure is in place to support FHIR Bulk Data for large programs, such as CMS clinical quality measures.

CMS may reduce provider burden by aligning values and value sets prescribed by CMS eCQMs with ontological standards and terminology used by providers. If prescribed values and value sets are not used by providers, providers would need to transform the data into appropriate format prior to delivery to CMS. This applies to HL7 FHIR and non-HL7 FHIR methods of submitting clinical quality data.

b. In what ways can the interoperability and quality reporting responsibilities of providers be consolidated so investments can be dually purposed?

If CMS adopts HL7 FHIR for clinical quality measures, CMS could consider collaborating with ASTP to test providers' Certified Health IT's HL7 FHIR server when each provider submits their CMS clinical quality measure data, assuming CMS clinical quality measures cover a wide breadth of HL7 FHIR resources. Additionally, this could provide future incentive opportunities for providers to support their digital health product pursuits. While ASTP oversees the certification of Certified Health IT via its Real-World Testing program, the uniqueness of each Certified Health IT instance may require additional testing of Certified Health IT at the instance level and not at the product level. The proposed CMS and ASTP effort is a potential solution.

c. Are there requirements CMS should consider for data registries to support digital quality measurement in a more efficient manner? Are there requirements CMS should consider for data registries that would support access to real-time quality data for healthcare providers to inform clinical care in addition to simplifying reporting processes?

CMS could consider automation and where it may be implemented within the current clinical quality measure reporting workflow and framework. APIs and HL7 FHIR support near real-time, if not real-time, data exchange. Additionally, APIs are often used to support predictable, automated workflows.

4. Information Blocking

PR-14. How can CMS encourage providers to submit information blocking complaints to ASTP/ONC's Information Blocking Portal? What would be the impact? Would it advance or negatively impact data exchange?

Increase visibility to the ASTP/ONC Information Blocking Portal. ASTP/ONC should increase visibility and navigability to the Information Blocking Portal. ASTP/ONC and CMS should consider additional awareness programs, as many providers may not be aware of Information Blocking.

D. Payers

3. Technical Standards and Certification

TD-4. How can CMS better encourage use of open, standards-based, publicly available APIs over proprietary APIs?

Identify and develop business use cases for APIs and HL7 FHIR interoperability. There is a need to identify ways to either monetize interoperability via open, standards-based APIs and HL7 FHIR or present how open, standards-based APIs and HL7 FHIR are saving money (e.g., reducing duplicative data collection, reducing data processing, reducing data storage, increasing time saved clinically and administratively). Providers are required to invest capital in Certified Health IT that support APIs and HL7 FHIR (Certified Health IT certified to 170.315(g)(10)). These are significant costs that may not be affordable for some provider demographics, such as rural or FQHCs. Thus, there is a need to identify business use cases for interoperability via open, standards-based, publicly available APIs and HL7 FHIR.

Expand required data elements within USCDI. CMS could collaborate with ASTP to expand the number of required data elements in USCDI. This would ensure that USCDI covers a broader range of clinical, administrative, and billing use cases and workflows. Additionally, ASTP could collaborate with CDC and NIH to ensure that USCDI covers public and population health and research, respectively. Alternatively, if there is the ability to codify USCDI+ within regulation as a requirement within 170.315(g)(10), ASTP could leverage USCDI+ domains to ensure that all data relevant to healthcare, including clinical, administrative, billing, public and population health, and research, are accessible and exchangeable via open, standards-based, publicly available APIs.

Develop online communities to spread awareness of standards-based, publicly available APIs. CMS and ASTP should create and nurture communities both in-person (HL7 FHIR Connectathons) and virtually (e.g., HL7 FHIR Zulip). This increases awareness of open, standards-based, publicly available APIs, develops the workforce, and promotes participation within the greater HL7 FHIR community.

Improve navigability to open-source resources and documentation. Currently, it is difficult to find and navigate to available open-source HL7 FHIR resources (e.g., test environments, test data, HL7 FHIR validators) and documentation (e.g., HL7 FHIR implementation guides) if one was not already aware. By improving navigation to these critical tools, the barrier to entry for developers to use open, standards-based, publicly available APIs is decreased.

TD-5. How could a nationwide provider directory of FHIR endpoints improve access to health information for patients, providers, and payers? Who should publish such a directory, and should users bear a cost?

For HL7 FHIR, a nationwide provider directory of FHIR endpoints would serve as critical infrastructure for the implementation and adoption of HL7 FHIR, since HL7 FHIR endpoints are how external data users access health information within Certified Health IT, using APIs and HL7 FHIR, at provider sites.

A national directory of provider HL7 FHIR endpoints should not be a cost burden to providers, as it would only be an additional barrier for HL7 FHIR users. CMS and/or ASTP should investigate if it is feasible to develop a nationwide provider directory of FHIR endpoints. In addition to being a resource to HL7 FHIR users, CMS and ASTP would both benefit from having a nationwide provider directory of FHIR endpoints. In the future, CMS and ASTP could use the nationwide provider directory of FHIR endpoints to query for program metrics and other reporting needs.

TD-7. To what degree has USCDI improved interoperability and exchange and what are its limitations?

d. Given improvements in language models, would you prefer a non-proprietary but less structured format that might improve data coverage even if it requires more processing by the receiver?

Clinovations supports defining the "what" of data exchange through clear, structured expectations, alongside preferred standards for implementation. Like in other industries—banking, travel, and finance—core use cases should be served by highly standardized, structured, and sometimes proprietary formats. We recommend that this approach apply to both USCDI and USCDI+, especially for near-term, well-defined use cases.

Today, USCDI's support for narrative notes is primarily limited to discharge summaries, leaving out a broad range of clinically relevant documentation. This gap presents a significant barrier to comprehensive information exchange. A non-proprietary approach to exchanging unstructured data—such as nursing, surgical, procedure, and labor and delivery notes—could enable broader utility and downstream processing by tools such as large language models (LLMs) and other AI methods. Such data elements are increasingly essential for real-time decision-making, risk stratification, and quality measurement.

To support this vision, the federal government should fund pilots that compare the burden and outcomes of traditional structured/standards-based exchange with more flexible approaches that leverage receiver-side capabilities to process unstructured data.

As real-world implementers, the Clinovations team frequently supports provider organizations in deploying health IT and enabling data exchange. In practice, even when vendors technically support standards like FHIR, the burden of mapping local data to FHIR resources falls squarely on provider organizations. While FHIR servers may exist, only a subset of data—typically tied to predefined vendor workflows—is exposed. Longstanding EHR implementations often lack mappings for large volumes of relevant data.

Our clients commonly rely on platforms such as OMOP, Databricks, and Snowflake to examine the native data landscape and identify elements that could be mapped to FHIR resources. However, this mapping effort remains costly, complex, and duplicative across organizations. Even when modern EHRs publicly support a wide array of FHIR resources, the data returned from queries is often limited to defaults that have been pre-mapped.

To enable meaningful interoperability at scale, CMS and ONC must address this persistent gap between theoretical API capability and practical data access. Some approaches could include:

- Employ a tiered model for a standards hierarchy: base USCDI/USCDI+ supported via standardized formats; extended/unstructured content supported via flexible formats and AI-ready payloads.
- Update the ONC Health IT Certification Program: Require vendors and implementers to document FHIR mappings across all USCDI elements, including narrative notes.
- Incentivize or support data mapping efforts: CMS could consider value-based care payment incentives or technical assistance programs to support organizations that go beyond default mappings and publish and share their approaches and mappings. Smaller organizations and those that do not have commercial payer customers do not have the same resources as larger organizations with in-house health informatics expertise.
- Focused Insights/Reporting: CMS could require vendors to publish real-time statistics on which FHIR resources and data elements are actively mapped and accessible in production environments.

TD-8. What are the most effective certification criteria and standards under the ONC Health IT Certification Program?

With the modernization and evolution of digital health and data systems, Clinovations sees new models and solutions in the market. While it made sense to sunset many criteria that were ubiquitous and established within EHRs, CMS and ASTP/ONC should consider creating an inventory of sunset criteria that must be supported via attestation. This will support future innovations in including capabilities that were previously supported by the Certification Program.

The most effective criteria and standards in recent years have focused on interoperable information exchange. As implied in TD-9, prioritizing API-enabled capabilities over software functionality would support innovation and not overburden any singular technology platform. With advances and increased adoption in AI/Ambient Scribe capabilities, narrative and unstructured data can be captured and stored more easily, reducing dependence on software functionality (TD-9a).

TD-9. Regarding certification of health IT:

e. How could EHRs capable of bulk data transfer be used to reduce the burden on providers for reporting quality performance data to CMS? What capabilities are needed to show benefit? What concerns are there with this approach?

Please refer to response to TD-8 regarding prioritizing of exchange and API capabilities vs. software functionality. Please see additional perspective regarding TD-9(d), below:

The framing of this question suggests that providers are unmotivated to respond to API-based data requests. However, as outlined in our response to TD-7, the underlying issue lies not in provider willingness, but in the technical and operational burden imposed on providers and vendors to map legacy data to modern FHIR resources. Most current EHR systems in active use today are not inherently FHIR-native. While there are promising developments among newer entrants to the market, the

majority of enterprise EHRs were implemented more than a decade ago and are highly customized to local workflows.

Vendors are responsible for standing up FHIR APIs and mapping core data to FHIR resources, but even when these APIs are technically compliant, the data exposed is typically limited to default mappings. The flexible architecture of mature EHR platforms makes data harmonization challenging, and customizing FHIR mappings for local configurations is labor-intensive and costly. This burden often falls on provider organizations—particularly those without in-house technical resources or robust vendor support.

To address this gap, CMS and ASTP should promote alternative data access strategies that shift mapping responsibilities away from individual providers. For example, enabling structured data extracts that intermediaries or receivers can process and map to FHIR or other standardized API specifications would distribute effort more equitably and accelerate real-world data exchange.

We were disappointed to see the Insights Conditions measures removed from upcoming policy proposals. These metrics had the potential to address long-standing bipartisan requests from Congress for transparency on data availability, interoperability, and exchange. The absence of these measures limits the ability to identify and improve performance bottlenecks across vendors and provider organizations. As the adage goes, "you can't improve what you can't measure." Today, we lack even basic visibility into API request volumes, response completeness, and the types of data most frequently requested or withheld. This opacity stifles progress.

We recommend that CMS and ASTP reconsider their measurement strategy by collaborating with providers, health system administrators, health IT developers, and other measurement stakeholders to pursue a co-created set of measures that balance industry burden with a measurement strategy that follows a baseline of consistent measurement approaches across reporting bodies. Lightweight, scalable exchange metrics could include self-reported measures that reduce implementation burden, while being carefully designed to preserve comparability across systems. Where possible, exchange-related metrics should be tied not only to EHR vendors and implementers but also to Health Information Exchanges (HIEs) and entities participating in TEFCA.

Some considerations for CMS and ASTP include:

- Reinforce "non-FHIR-native" reality: EHR vendor FHIR support is limited in production and varies greatly – enterprise vendors offer more capabilities than ambulatory only or vendors that typically serve smaller hospitals or ambulatory practices.
- Reconsider API usage metrics: Recommend CMS establish a national dashboard or reporting repository that tracks API request types, volumes, and response quality (possibly leveraging TEFCA QHIN participation).
- Encourage incentives for transparency: Consider approaches to recognize vendors and implementers who voluntarily report exchange metrics.
- Promote Intermediary Ecosystems: ONC or CMS regulations and programs should support the real-world reality of third-party intermediaries (like clearinghouses or FHIR translation hubs) that can reduce the need for each provider to handle FHIR mappings independently.

TD-10. For EHR and other developers subject to the ONC Health IT Certification Program, what further steps should ASTP/ONC consider to implement the 21st Century Cures Act's API condition of certification (42 U.S.C. 300jj-11(c)(5)(D)(iv)) that requires a developer's APIs to allow health information to be accessed, exchanged, and used without special effort, including providing access to all data elements of a patient's electronic health record to the extent permissible under applicable privacy laws?

Expand required data elements within USCDI. CMS could consider collaborating with ASTP to expand the number of required data elements in USCDI. This would ensure that USCDI covers a broader range of clinical, administrative, and billing use cases and workflows. Additionally, ASTP could collaborate with CDC and NIH to ensure that USCDI covers public and population health and research, respectively. Alternatively, if there is the ability to codify USCDI+ within regulation as a requirement within 170.315(g)(10), ASTP could leverage USCDI+ domains to ensure that all data relevant to healthcare, including clinical, administrative, billing, public and population health, and research, are accessible and exchangeable via open, standards-based, publicly available APIs.

TD-11. As of January 1, 2024, many health IT developers with products certified through the ONC Health IT Certification Program are required to include the capability to perform an electronic health information export or "EHI export" for a single patient as well as for patient populations (45 CFR 170.315(b)(10)). Such health IT developers are also required to publicly describe the format of the EHI export. Notably, how EHI export was accomplished was left entirely to the health IT developer. Now that this capability has been in production for over a year, CMS and ASTP/ONC seek input on the following:

c. Should CMS consider policy changes to support this capability's use?

Please refer to Clinovations response to PR-7 which specifically addresses EHI Export.

4. Data Exchange

TD-13. What new opportunities and advancements could emerge with APIs providing access to the entirety of a patient's electronic health information (EHI)?

Clinical – claims linked data. Clinical – claims linked data is sought after data that is often considered completer and more useful compared to clinical data or claims data.

Expansion of data elements exchangeable with digital health products. APIs providing access to a full EHI dataset would broaden the third-party mobile app market, and the digital health product market overall, as it would enable the development of products for a wider arrange of use cases and workflows.

Expansion of data elements for the development of AI tools and dashboards. APIs providing access to a full EHI dataset would expand usable data for AI tools and dashboards.

a. What are the primary obstacles to this?

Capital and resources to support development and implementation. Developers of Certified Health IT and providers will need to spend additional capital to further develop and update, respectively, their systems and APIs.

Heterogeneous data formats. Data formats vary, depending on the data element within the full EHI. The full EHI will contain not only information contained with USCDI, but also additional data elements including but not limited to health plans, payment, and explanation of benefits. The data format within these IT systems that house these additional data elements may not be supported by Certified Health IT. CMS and/or ASTP should investigate the feasibility of developing APIs that would enable users to access a patient's full EHI.

Support of data elements within HL7 FHIR. CMS should investigate and verify that HL7 FHIR R4 supports all data elements within a patient's full EHI.

b. What are the primary tradeoffs between USCDI and full EHI, especially given more flexible data processing capabilities today?

A full EHI would have more information, compared to only USCDI, since USCDI contains minimal information related to enrollment, payment, claims adjudication, and case or medical management record systems maintained by or for a health plan. However, this would increase the number of USCDI data elements, increasing burden on developers of Certified Health IT and providers. When considering this value proposition, it will be critical to understand the level of effort required to develop, implement, collect and maintain the full EHI compared to the perceived value by the end data user.

TD-15. Regarding bulk FHIR APIs:

a. How would increased use of bulk FHIR improve use cases and data flow?

Support automation within reporting programs. FHIR Bulk Data APIs can be leveraged to pull data on a predictable, periodic basis from reporting sites. Future opportunities may also include the feeding of this data into a metrics calculator to further automate programs.

Support in third-party applications. While there are currently few use cases for patient use of the HL7 FHIR Bulk Data Access implementation guide (IG), it is feasible that a patient may use it, given that they a) have a commercial product (e.g., a third-party mobile app) or b) built a product (e.g., a third-party app) that supports the HL7 FHIR Bulk Data Access IG. Similarly non-provider data users may also benefit from increased use of HL7 FHIR Bulk Data, including but not limited to public health registries, payers, labs, vital records offices, and HIEs/National Networks/QHINs.

b. What are the potential disadvantages of their use?

No real-time data exchange. Currently, the HL7 FHIR Bulk Data Access IG V2.0.0 does not support real-time data exchange.

Necessitates a FHIR Bulk Data Output File Server. The HL7 FHIR Bulk Data Access IG V2.0.0 requires the use of a FHIR Bulk Data Output File Server, requiring an additional server, compared to needing only a FHIR Authorization Server and a FHIR Resource Server.

Increases server processing and computing. Large HL7 FHIR Bulk Data Access queries may result in server errors due to the increase load on servers.

TD-16. What are the tradeoffs of maintaining point-to-point models vs. shared network infrastructure?

Point-to-point models support real-time data exchange whereas shared network infrastructure will not be, since the data will need to be routed through the intermediary in between the data sender and the end data receiver.

Amongst a variety of roles it serves, the intermediary serves as a directory of FHIR endpoints for providers connected to the shared network infrastructure and as a record locator service. Without a nationwide provider directory of FHIR endpoints, it is sometimes challenging to find provider FHIR endpoints, especially if it is not publicly available.

It is more burdensome to pull from multiple providers within a point-to-point model, compared to a shared network infrastructure. Since data is pulled on a period basis in a shared network infrastructure (e.g., an HIE) into an intermediary repository, a patient's medical history may be pulled together in one query, even if the source of the data was from different providers. In a point-to-point model, one would need to query each provider FHIR endpoint, individually, instead of once against an intermediary repository to obtain the same data.

a. Do current rules encourage scalable network participation?

In a shared network infrastructure, ideally there is one network that encompasses the entirety of the United States. Otherwise, data users will be querying against multiple shared network infrastructures, which is still an improvement compared to the point-to-point model. TEFCA appears to be the beginnings of a singular shared network. However, more work needs to be done to connect remaining unconnected providers and networks. Although CMS and/or ASTP would need to investigate more, it may be more beneficial for provider to connect directly, instead of connecting through networks, since providers who are connecting to TEFCA would have an additional intermediary.

b. What changes would improve alignment (for example, API unification, reciprocal access)?

If or when a nationwide provider directory of FHIR endpoints is available, point-to-point data exchange will be more feasible, requiring the intermediary within a shared network infrastructure to shift roles. In this scenario, the intermediary has an opportunity to shift to a "translator" role, transforming data from legacy formats to HL7 FHIR.

TD-17. Given operational costs, what role should CMS or ASTP/ONC or both have in ensuring viability of healthcare data sharing networks, including enough supply and demand, that results in usage and outcomes?

Identify and develop business use cases for APIs and HL7 FHIR interoperability. There is a need to identify ways to either monetize interoperability via open, standards-based APIs and HL7 FHIR or present how open, standards-based APIs and HL7 FHIR are saving money (e.g., reducing duplicative data collection, reducing data processing, reducing data storage, increasing time saved clinically and administratively). Providers are required to invest capital in Certified Health IT that support APIs and HL7 FHIR (Certified Health IT certified to 170.315(g)(10)). These are significant costs that may not be affordable for some provider demographics, such as rural or FQHCs. Thus, there is a need to identify business use cases for interoperability via open, standards-based, publicly available APIs and HL7 FHIR.